

Symbol	Unit	Meaning		
<i>Indices</i>				
i, o	Token	Input (sold) and output (bought) endpoint tokens, respectively; (i, o) is an ordered input-output pair and the direct route is $i \rightarrow o$.		
x	Token	Generic token index where a quantity applies to an endpoint or candidate token.		
k	Token	Candidate vehicle token used as the route intermediate; $k \notin \{i, o\}$.		
h	Token	Candidate vehicle token used as a challenger to incumbent k ; $h \in \mathcal{K} \setminus \{i, o, k\}$.		
ℓ, p, p'	Token / pool	ℓ indexes tokens in cross-token sums; p indexes a focal liquidity pool; p' indexes another pool in leave-one-out sums.		
a, a'	LP address	a indexes a focal liquidity-provider address or resolved position controller; a' indexes another provider in within-pool sums.		
b	Fraction	Symmetric price-band half-width used to measure concentrated liquidity around the current reference price; $b = 0.02$ is primary.		
t, u, w	UTC day / UTC week	t indexes a focal calendar day; u indexes another calendar day in a time-window sum; w indexes calendar weeks in settlement variables.		
τ	Days	Positive calendar-day horizon selected ex ante for each dynamic specification.		
d, μ	Calendar days / calendar months	d is calendar-day event time relative to a candidate-specific shock; μ is calendar-month event time relative to an architecture activation date. Event time zero contains the event or activation date.		
g	Settlement comparison cell	Settlement comparison cell defined by ordered pair (i, o) , vehicle k , UTC week w , and a prespecified route-size bin.		
q	USD	Input quote notional. Un-suffixed route-cost columns use $q = \$10,000$.		
r	Route unit	Reconstructed input-to-output route-unit. A coherent $i \rightarrow k \rightarrow o$ component contributes one r regardless of its number of legs; a split or join contributes one r per reconstructed input-output pair.		
<i>Route and liquidity aggregates</i>				
$\text{Vol}_t, \text{DVol}_t$	USD	Vol_t is total realized USD volume across all direct and indirect route units on day t . DVol_t restricts it to direct routes, so $0 \leq \text{DVol}_t \leq \text{Vol}_t$.		
$\text{IVol}_t, \text{IVol}_{k,t}$	USD	IVol_t restricts Vol_t to indirect routes, so $0 \leq \text{IVol}_t \leq \text{Vol}_t$. $\text{IVol}_{k,t}$ further restricts that volume to routes using vehicle k , so $0 \leq \text{IVol}_{k,t} \leq \text{IVol}_t$.		
$\text{Vol}_{i,o,t}, \text{IVol}_{i,o,t}, \text{IVol}_{i,o,k,t}$	USD	$\text{Vol}_{i,o,t}$ is realized route-unit volume for ordered pair (i, o) on day t . $\text{IVol}_{i,o,t}$ restricts it to indirect routes, and $\text{IVol}_{i,o,k,t}$ further restricts it to routes through k , so $0 \leq \text{IVol}_{i,o,k,t} \leq \text{IVol}_{i,o,t} \leq \text{Vol}_{i,o,t}$.		
$N_t, N_{k,t}^{\text{in}}, N_{k,t}^{\text{out}}$	Route-unit count	N_t counts all reconstructed route units r on day t , not their individual legs. $N_{k,t}^{\text{in}}$ and $N_{k,t}^{\text{out}}$ restricts to routes with k as the input or output endpoint, respectively; each lies between 0 and N_t .		
$N_t^i, N_{k,t}^i$	Route-unit count	N_t^i restricts N_t to route units with at least one intermediate token. $N_{k,t}^i$ further restricts that count to route units k , so $0 \leq N_{k,t}^i \leq N_t^i \leq N_t$; superscript i denotes i direct routes.		
$\mathcal{A}_t, \mathcal{A}_t^k, \mathcal{M}_t^k$	Sets of token pairs	$\mathcal{A}_t$ is the set of endpoint pairs active in indirect routes on day t . $\mathcal{A}_t^k$ restricts it to pairs using k , and $\mathcal{M}_t^k$ further restricts it to pairs for which k has the largest candidate-vehicle volume: $\mathcal{M}_t^k \subseteq \mathcal{A}_t^k \subseteq \mathcal{A}_t$.		
$\text{LogVol}_{k,t}^{\text{in}}, \text{LogVol}_{k,t}^{\text{out}}$	USD	$\text{LogVol}_{k,t}^{\text{in}}$ and $\text{LogVol}_{k,t}^{\text{out}}$ are route-leg USD volumes for which k is the leg's input or output token, respectively, on day t .		
$\text{Vol}_{k,t}^{\text{in}}, \text{Vol}_{k,t}^{\text{out}}$	USD	Input- and output-endpoint restrictions of Vol_t for token k : $0 \leq \text{Vol}_{k,t}^{\text{in}} \leq \text{Vol}_t$ and $0 \leq \text{Vol}_{k,t}^{\text{out}} \leq \text{Vol}_t$.		
$\mathcal{K}$	Set of tokens	Prespecified candidate set $\{\text{WETH}, \text{USDC}, \text{USDT}, \text{DAI}, \text{WBTC}\}$.		
$\mathcal{L}_t, \mathcal{L}_{k,t}, m_p$	Set of pools / token count	$\mathcal{L}_t$ contains valid Uniswap V3 daily-snapshot pools with exact token contents identified in the persisted V3 swap archive and $0 < \text{TVL}_{k,t} \leq 10$ billion USD. $\mathcal{L}_{k,t} \subseteq \mathcal{L}_t$ restricts that set to pools with candidate k on one side. m_p is the number of pool tokens in $\mathcal{K}$, so $m_p \in \{1, 2\}$.		
$\text{TVL}_{p,t}, L_{k,t}$	USD	$\text{TVL}_{p,t}$ is pool p 's day- t USD TVL from the V3 daily snapshot. $L_{k,t} = \sum_{p \in \mathcal{K}_k} \text{TVL}_{p,t} / m_p$; a one-candidate pool contributes all TVL, to that candidate and a two-candidate pool contributes one half to each.		
$\text{PoolVol}_{p,t}, \text{SpokeVol}_{p,k,t}$	USD	$\text{PoolVol}_{p,t}$ is total realized swap volume in pool p on day t . $\text{SpokeVol}_{p,k,t}$ restricts it to pool legs belonging to reconstructed indirect routes that use k as the intermediate, so $0 \leq \text{SpokeVol}_{p,k,t} \leq \text{PoolVol}_{p,t}$.		
<i>LP portfolio objects</i>				
$L_{a,p,t}$	USD	End-of-day USD value of active liquidity supplied by provider a to pool p , marked at the day- t reference price.		
$w_{a,p,t}$	Fraction (0–1)	Provider a 's share of active capital in pool p on day t : $w_{a,p,t} = L_{a,p,t} / \sum_{a'} L_{a',p,t}$.		
$F_{a,p,t}^{\text{LP}}$	USD per day	Day- t liquidity deposits by a into p minus withdrawals, valued at transaction-time USD prices and excluding fee collections.		
$\text{Fee}_{a,p,t}, C_{a,p,t}^{\text{LP}}$	USD	Swap fees accrued to provider a in pool p during day t , and USD gas spent by a on that position's mid-order, fee , gas , and $\text{fee} + \text{gas}$ transactions.		
$V_{a,p,t}^{\text{LP}}, V_{a,p,t}^{\text{RB}}$	USD	End-of-day values of the day- t opening LP inventory and its self-financing rebalanced benchmark, respectively, both excluding day- t fees, flows, and gas.		
$R_{a,p,t}^{\text{LP}}, R_{a,-p,t}^{\text{other}}$	Daily return fraction	$R_{a,p,t}^{\text{LP}} = (V_{a,p,t}^{\text{LP}} - L_{a,p,t-1}) / L_{a,p,t-1}$ is the position's fee-, flow-, and gas-excluded return. $R_{a,-p,t}^{\text{other}}$ is the lag-capital-weighted mean of that return across a 's pools other than p .		
$\omega_{a,-,-p,t}$	Fraction (0–1)	Share of provider a 's marked LP inventory outside focal pool p exposed to token x , excluding both tokens in p and normalized to sum to one across the remaining tokens.		
$\text{BandDepth}_{p,t,b}$	USD	Average across trade directions of the input USD notional executable in pool p before its marginal price exits the symmetric b band around the day- t reference price.		
<i>Route and liquidity aggregates</i>				
R_t^{WETH}	Daily log-return fraction	Day- t log return of the WETH price used to construct downside stress.		
<i>Candidate-shock objects</i>				
$P_{k,t}, P_{k,t}^{\text{date}}$	USD per token	$P_{k,t}$ is the day- t USD price of any token x . $P_{k,t}^{\text{date}}$ restricts that price to candidate $k \in \mathcal{K}$ on day t .		
$R_{k,t}, R_{k,t}^{\text{date}}$	Daily log-return fraction	$R_{k,t} = \ln(P_{k,t} / P_{k,t-1})$ is the day- t return of any token x . $R_{k,t}^{\text{date}}$ restricts it to candidate $k \in \mathcal{K}$.		
$\sigma_{k,t-1}^{\text{SD}}$	Daily log-return standard deviation	Sample standard deviation of $R_{k,u}$ over the 30 calendar days ending at $t - 1$, requiring at least 20 valid daily returns.		
<i>Architecture objects</i>				
$R_{i,o,t}, \sigma_{i,o}^{\text{pre}}$	Daily log-return fraction / daily log-return standard deviation	$R_{i,o,t} = \ln((P_{i,t} / P_{o,t}) / (P_{i,t-1} / P_{o,t-1}))$ is the ordered endpoint-pair return. $\sigma_{i,o}^{\text{pre}}$ is its sample standard deviation over $\mathcal{T}_{i,o}^{\text{V3}}$ activity.		
<i>Persistence and displacement objects</i>				
$k_{i,o,t}^*, h_{i,o,q,t}^*$	Token	$k_{i,o,t}^*$ is the incumbent vehicle with the largest mean $\text{VehicleShare}_{i,o,k,u}$ over the 30 calendar days ending at $t - 1$. $h_{i,o,q,t}^*$ is the executable nonincumbent candidate with the smallest all-in indirect cost $C_{i,o,h,q,t}^*$ on day t .		
<i>Architecture objects</i>				
$t_0^{\text{V3}}, \mathcal{T}_{\text{pre}}^{\text{V3}}, \mathcal{P}_q^{\text{V3}}$	UTC day / set of UTC days / set of token pairs	t_0^{V3} is the Uniswap V3 activation date, 5 May 2021. $\mathcal{T}_{\text{pre}}^{\text{V3}}$ is the fixed 180-calendar-day window ending at $t_0^{\text{V3}} - 1$. $\mathcal{P}_q^{\text{V3}}$ is the fixed architecture-analysis universe of q token pairs that have realized route volume on at least 30 days in that pre-period; every member is quoted at q throughout the event window, independent of post-V3 activity.		
$t_0^{\text{V4}}, \mathcal{T}_{\text{pre}}^{\text{V4}}, \mathcal{T}_{i,o,\text{pre}}^{\text{V4}}$	UTC day / sets of UTC days	t_0^{V4} is the Ethereum V4 activation date established from the canonical deployment metadata before extraction. $\mathcal{T}_{\text{pre}}^{\text{V4}}$ is the fixed 180-calendar-day window ending at $t_0^{\text{V4}} - 1$. $\mathcal{T}_{i,o,\text{pre}}^{\text{V4}} \subseteq \mathcal{T}_{\text{pre}}^{\text{V4}}$ restricts that window to days with $\text{Vol}_{i,o,t} > 0$.		
<i>Route-cost quote objects</i>				
$\mathcal{P}_{k,t,q}$	Set of token pairs	Quote-universe pairs for candidate k : distinct ordered input-output pairs among day t 's 200 largest reconstructed pairs by realized USD volume, where clean means route tokens are single or coherent, $k \notin \{i, o\}$, and each of i, o , and k has a valid day-price estimate. A valid estimate requires at least three finite token-side USD-per-token observations in $(0, \$1,000,000)$ equals their realized-USD-volume-weighted median. Each pair is submitted to the direct and via- k quote engines at input q ; membership does not require either quote to execute.		
$\mathcal{D}_{k,t,q}, \mathcal{Z}_{k,t,q}$	Sets of token pairs	$\mathcal{D}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ and $\mathcal{Z}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ restrict the broad pair set to pairs with an executable direct route or an executable indirect route via k .		
$\mathcal{C}_{k,t,q}$	Set of token pairs	Common-support pairs for candidate k , day t , and notional q : $\mathcal{C}_{k,t,q} = \mathcal{D}_{k,t,q} \cap \mathcal{Z}_{k,t,q}$, so both the direct and indirect routes are executable.		
$\mathcal{T}_{k,t,q}, \mathcal{W}_{k,t,q}$	Sets of token pairs	$\mathcal{T}_{k,t,q} \subseteq \mathcal{D}_{k,t,q}$ is the thin-direct subset. $\mathcal{W}_{k,t,q} \subseteq \mathcal{C}_{k,t,q}$ contains common-support pairs for which the indirect route beats the direct route, equivalently $\Delta C_{k,t,q}^D < 0$.		
$D_{i,o,q,t}, I_{i,o,h,q,t}, T_{i,o,q,t}$	Indicator (0/1)	Indicators for direct-route availability (D), indirect-route-through- k availability (I), and an executable direct route returning less than $0.9q$ (T).		
$O_{i,o,q,t}^D, O_{i,o,h,q,t}^I$	USD	Quoted output values; superscripts D and I denote direct and indirect routes.		
$O_{i,o,q,t}^{D,\text{fee}}, O_{i,o,h,q,t}^{I,\text{fee}}$	USD	Counterfactual direct and indirect output values when each pool remains at its pre-quote marginal price but charges its historical swap fee; these retain fee loss and suppress price impact.		
$C_{i,o,q,t}^D, C_{i,o,h,q,t}^I$	USD	Historical gas expenditure for the quoted direct and indirect routes, respectively, using route-specific gas usage and the day- t gas-linked USD price.		
$C_{i,o,q,t}^{D^*}, C_{i,o,h,q,t}^{I^*}$	Fraction	All-in execution-cost fractions for the direct and indirect alternatives, including quoted pool fees and price impact plus historical gas expenditure.		
$\Delta C_{i,o,k,q,t}^{D^*}$	Fraction	Pair-level direct cost advantage $(O_{i,o,q,t}^D - O_{i,o,h,q,t}^I) / O_{i,o,q,t}^D$, on $\mathcal{C}_{k,t,q}$; positive values favor the direct route.		
$\Delta C_{i,o,k,q,t}^{D^{\text{all}}}$	Fraction	All-in direct cost advantage $C_{i,o,h,q,t}^* - C_{i,o,q,t}^*$ on $\mathcal{C}_{k,t,q}$; positive values favor the direct route.		
<i>Settlement objects and operators</i>				
$\mathcal{R}_{k,w}, \mathcal{R}_{k,w}^{\text{transfer}}$	Set of route units Set of route units	Receipt-audited matched route units using vehicle k in UTC week w . Members whose receipt logs a transfer of vehicle k ;		
$\mathcal{R}_g^3, \mathcal{R}_g^4$	Sets of route units	Receipt-audited route units in settlement comparison cell g executed on Uniswap V3 and V4, respectively. A matched cell has both sets nonempty.		
$M_{r,k}, \text{GrossLegVol}_{r,k}$	USD	$M_{r,k}$ is the sum of absolute, route-attributed ERC-20 transfers of intermediate k between distinct nonzero addresses in route unit r , valued in USD. $\text{GrossLegVol}_{r,k}$ is the corresponding gross USD notional of k entering or leaving the route's pools across both vehicle legs.		
$ \mathcal{S} , \mathbf{1}_{\{\cdot\}}$	Count / indicator (0/1)	Cardinality of any finite set S and an indicator equal to one when its subscripted condition is true.		
<i>Dynamic operators</i>				
Δ_τ		Change in a daily variable over the τ days ending at t : $\Delta_\tau X_t = X_t - X_{t-\tau}$.		
Variable	Formula	Unit	Data column	Definition
<i>Vehicle-use measures</i>				
$\text{VehicleShare}_{k,t}$	$\frac{\text{IVol}_{k,t}}{\text{Vol}_t}$	Fraction (0–1)	bridge.share	Fraction of day- t indirect-route USD volume routed through k .
$\text{AllRouteVehicleShare}_{k,t}$	$\frac{\text{IVol}_{k,t}}{\text{Vol}_t}$	Fraction (0–1)	all_route.bridge.share	Fraction of day- t all-route USD volume routed indirectly through k .
$\text{VehicleCountShare}_{k,t}$	$\frac{N_{k,t}^i}{N_t^i}$	Fraction (0–1)	bridge.count.share	Fraction of day- t indirect route units that use k .
$\text{PairCoverage}_{k,t}$	$\frac{ \mathcal{A}_t^k }{ \mathcal{A}_t }$	Fraction (0–1)	pair.coverage	Fraction of active indirect-route endpoint pairs that use k .
$\text{MainVehiclePairShare}_{k,t}$	$\frac{ \mathcal{M}_t^k }{ \mathcal{A}_t }$	Fraction (0–1)	pair.main.vehicle.share	Fraction of active indirect-route endpoint pairs for which k has the largest candidate-vehicle volume.
$\text{IVol}_{k,t}$		USD	bridge.volume.usd	Sum of realized USD volume over indirect routes using vehicle k on day t .
<i>Pair-level route-choice measures</i>				
$\text{VehicleShare}_{i,o,h,t}$	$\frac{\text{IVol}_{i,o,k,t}}{\text{IVol}_{i,o,t}}$	Fraction (0–1)	actual.vehicle.share	Candidate k 's share of realized indirect-route USD volume for ordered pair (i, o) on day t , defined when $\text{IVol}_{i,o,t} > 0$.
$\text{IndirectRouteShare}_{i,o,t}$	$\frac{\text{IVol}_{i,o,t}}{\text{Vol}_{i,o,t}}$	Fraction (0–1)	pair.indirect.route.share	Fraction of realized route-unit USD volume executed through an indirect route on day t , defined when $\text{Vol}_{i,o,t} > 0$.
$\text{Coverage}_{i,o,t}^C$	$\sum_{k \in \mathcal{K} \setminus \{i,o\}} \text{VehicleShare}_{i,o,k,t}$	Fraction (0–1)	pair.candidate.vehicle.coverage	Share of pair (i, o) 's realized indirect-route volume routed through a candidate in $\mathcal{K} \setminus \{i, o\}$; defined when total pair indirect volume is positive.
$\text{VehicleHHI}_{i,o,t}$	$\sum_{k \in \mathcal{K} \setminus \{i,o\}} \left(\frac{\text{VehicleShare}_{i,o,k,t}}{\text{Coverage}_{i,o,t}^C} \right)^2$	Fraction (0–1)	pair.vehicle.hhi	Herfindahl concentration among candidates in $\mathcal{K} \setminus \{i, o\}$ for ordered pair (i, o) on day t , after normalizing their shares to sum to one defined when candidate-routed indirect volume is positive.
<i>Network and route-denominator controls</i>				
$\text{VolShare}_{k,t}$	$\frac{\text{LogVol}_{k,t}^{\text{in}} + \text{LogVol}_{k,t}^{\text{out}}}{\sum_{i \in \mathcal{K}} (\text{LogVol}_{i,t}^{\text{in}} + \text{LogVol}_{i,t}^{\text{out}})}$	Fraction (0–1)	vol.share	$\text{LogVol}_{k,t}^{\text{in}}$ and $\text{LogVol}_{k,t}^{\text{out}}$ are input-side and output-side route-leg USD volumes for k ; t indexes every token in the day- t network.
$\text{Betweenness}_{k,t}$	$\frac{N_{k,t}^i}{N_t - N_{k,t}^{\text{in}} - N_{k,t}^{\text{out}}}$	Fraction (0–1)	betweenness.centralitiy	Fraction of day- t intent routes on which k is intermediate, excluding routes where k is the input or output endpoint. Superscripts identify each route node.
$\text{Betweenness}_{k,t}^{\text{vol}}$	$\frac{\text{IVol}_{k,t}}{\text{Vol}_t - \text{Vol}_{i,o,t}^{\text{in}} - \text{Vol}_{i,o,t}^{\text{out}}}$	Fraction (0–1)	volume.weighted.betweenness	USD-volume analogue of $\text{Betweenness}_{k,t}$ using the same route universe and input/output-endpoint exclusions.
Vol_t	$\text{DVol}_t + \text{IVol}_t$	USD	daily.all.route.volume.usd	Total realized USD volume across direct and indirect route units on day t .
IVol_t		USD	daily.indirect.route.volume.usd	Total realized USD volume across indirect route units on day t .
$\text{IndirectRouteShare}_t$	$\frac{\text{IVol}_t}{\text{Vol}_t}$	Fraction (0–1)	indirect.route.share	Fraction of day- t all-route USD volume executed through indirect routes.
<i>Liquidity measures</i>				
$L_{k,t}$	$\sum_{p \in \mathcal{K}_k} \frac{\text{TVL}_{p,t}}{m_p}$	USD	vehicle.linked.liquidity.usd	Candidate k 's allocated share of valid Uniswap V3 pool TVL; full TVL when k is the pool's only candidate token and one half when both pool tokens are candidates.
$\text{LPConc}_{k,t}$	$\frac{L_{k,t}}{\sum_{i \in \mathcal{K}} L_{i,t}}$	Fraction (0–1)	lp.concentration	Candidate k 's share of total vehicle-linked liquidity on day t .
$\text{LogVehicleLiquidity}_{k,t}$	$\ln(1 + L_{k,t})$	Natural-log points	log.vehicle.linked.liquidity	Natural log of one plus $L_{k,t}$ measured in USD.
<i>Liquidity commonality measures</i>				
$\text{VehicleLiquidityFactor}_{p,k,t}$	$\frac{\sum_{p' \in \mathcal{L}_{k,t} \setminus \{t\}} \Delta_1 \ln(\text{TVL}_{p',t})}{ \mathcal{L}_{k,t} \setminus \{p\} }$	Daily log-change points	vehicle.factor.loo	Leave-one-out mean daily log TVL change among other valid pools linked to candidate k ; requires at least three other pools.
$\text{MarketLiquidityFactor}_{p,t}$	$\frac{\sum_{p' \in \mathcal{L}_t \setminus \{p\}} \Delta_1 \ln(\text{TVL}_{p',t})}{ \mathcal{L}_t \setminus \{p\} }$	Daily log-change points	market.factor.loo	Leave-one-out mean daily log TVL change across all other valid pools in $\mathcal{L}_t$; market-wide control for the vehicle factor.
$\text{VehicleRouteShare}_{p,k,t}$	$\frac{\text{SpokeVol}_{p,k,t}}{\text{PoolVol}_{p,t}}$	Fraction (0–1)	pool.vehicle.route.share	Fraction of pool p 's day- t realized swap volume contributed by legs of indirect routes using k as the intermediate; defined for positive pool volume.
<i>LP capital and return measures</i>				
$L_{a,p,t}$		USD	lp.active.capital.usd	End-of-day active liquidity controlled by address a in pool p , reconstructed from position events and marked at day- t reference prices. Contract-managed positions are assigned to a controller only where the event trail permits look-through.
$w_{a,p,t}, p_{a,p,t}^{\text{LP}}$	$\frac{L_{a,p,t}}{\sum_{a'} L_{a',t}}$ $\frac{F_{a,p,t}^{\text{LP}}}{L_{a,p,t}}$	Fraction (0–1) USD per day	lp.pool.capital.share lp.net.flow.usd	Provider a 's share of reconstructed active capital in pool p on day t . Transaction-time USD value of day- t mint deposits less burn withdrawals for provider a in pool p , with fee collections excluded.
$\text{LPFeeYield}_{a,p,t}$	$\frac{\text{Fee}_{a,p,t}}{L_{a,p,t-1}}$	Daily fraction	lp.fee.yield	Day- t fees accrued to the position divided by lagged active capital; defined only for positive lagged capital.
$\text{LVR}_{a,p,t}$	$\frac{V_{a,p,t}^{\text{RB}} - V_{a,p,t}^{\text{LP}}}{L_{a,p,t-1}}$	Daily fraction	lp.lvr	Underperformance of the opening LP inventory relative to its self-financing rebalanced benchmark, before fees, flows, and gas, defined for positive lagged capital.
$\text{LPNetReturn}_{a,p,t}$	$\text{LPFeeYield}_{a,p,t} - \text{LVR}_{a,p,t}$	Daily fraction	lp.net.return	Fee yield less loss versus rebalancing and position-management gas cost, all scaled by lagged active capital.
$p_{a,-p,t}^{\text{other}}$	$\frac{\frac{G_{a,p,t}^{\text{LP}}}{L_{a,p,t-1}}}{\sum_{p' \neq p} \frac{L_{a,p',t-1} R_{a,p',t}^{\text{LP}}}{\sum_{p'' \neq p} L_{a,p'',t-1}}}$	Daily return fraction	lp.other.pool.return	Lag-capital-weighted, fee-, flow-, and gas-excluded return on provider a 's positions outside focal pool p ; defined when outside lagged capital is positive.
$\mathcal{Z}_{a,-p,t}^{\text{other}}$	$\sum_{a \notin p} \omega_{a,-,-p,t-1} R_{a,t}$	Daily return fraction	lp.predicted.other.pool.shock	Shift-share return shock formed from provider a 's lagged outside-pool token exposures and independent day- t token returns, excluding both focal-pool tokens.
$\text{LPWealthShock}_{p,t}$	$\sum_a w_{a,p,t-1} \mathcal{Z}_{a,-p,t}^{\text{other}}$	Daily return fraction	pool.lp.wealth.shock	Lagged-capital-share-weighted predicted outside-pool shock of the providers supplying pool p before day- t flows.
$\text{LPOverlap}_{p,p',t}$	$\sum_a \min\{w_{a,p,t}, w_{a,p',t}\}$	Fraction (0–1)	lp.provider.overlap	Common active-capital share attributable to the same resolved providers in pools p and p' ; zero means no overlap and one means identical provider weights.
<i>Route-cost opportunity measures</i>				
$\text{AnyIndirectAvailable}_{i,o,h,t}$	$\mathbf{1}_{\{k \in \mathcal{K} \setminus \{i,o\} : I_{i,o,h,q,t} > 0\}}$	Indicator (0/1)	any.indirect.available	Equals one when at least one candidate in $\mathcal{K} \setminus \{i, o\}$ provides an executable indirect route for pair (i, o) at input notional q .
$\text{DirectDepth}_{i,o,q,t}$	$\frac{O_{i,o,q,t}^D}{q}$	Output USD per input USD	pair.direct.depth	Direct-route output ratio for cells with $D_{i,o,q,t} = 1$; undefined when the direct route is not executable.
$\text{IndirectDepth}_{i,o,h,q,t}$	$\frac{O_{i,o,h,q,t}^I}{q}$	Output USD per input USD	pair.indirect.depth	Indirect-route output ratio through candidate k for $(i, o) \in \mathcal{Z}_{k,t,q}$; undefined when that route is not executable.
$C_{i,o,q,t}^{D,\text{fee}}$	$1 - \frac{O_{i,o,q,t}^{D,\text{fee}}}{q}$	Fraction	pair.direct.fee.cost	Direct-route loss attributable to historical pool fees when the direct impact is suppressed at the pre-quote marginal price.
$C_{i,o,q,t}^{D,\text{impact}}$	$\frac{O_{i,o,q,t}^{D,\text{fee}} - O_{i,o,q,t}^D}{q}$	Fraction	pair.direct.price.impact.cost	Additional direct-route output loss caused by moving through the realized liquidity curve, after holding the fee-only output fixed.
$C_{i,o,q,t}^{D,\text{gas}}$	$\frac{G_{i,o,q,t}^D}{q}$	Fraction	pair.direct.gas.cost	Historical direct-route gas expenditure as a fraction of input USD notional.
$C_{i,o,q,t}^{I,\text{fee}}$	$1 - \frac{O_{i,o,h,q,t}^{I,\text{fee}}}{q}$	Fraction	pair.indirect.fee.cost	Two-leg indirect-route loss attributable to both pools' historical fees when price impact is suppressed at pre-quote marginal price.
$C_{i,o,k,q,t}^{I,\text{impact}}$	$\frac{O_{i,o,h,q,t}^{I,\text{fee}} - O_{i,o,h,q,t}^I}{q}$	Fraction	pair.indirect.price.impact.cost	Additional two-leg indirect-route output loss caused by moving through both realized liquidity curves, after holding fee-only output fixed.
$C_{i,o,k,q,t}^{I,\text{gas}}$	$\frac{G_{i,o,h,q,t}^I}{q}$	Fraction	pair.indirect.gas.cost	Historical two-leg indirect-route gas expenditure as a fraction of input USD notional.
$C_{i,o,q,t}^D$	$1 - \frac{O_{i,o,q,t}^D}{q} + \frac{G_{i,o,q,t}^D}{q}$	Fraction	pair.all.in.direct.cost	Direct quote-output loss, which includes pool fees and price impact, plus historical route gas as a fraction of input notional. Fee, price-impact, and gas contributions are retained separately and sum to this measure.
$C_{i,o,k,q,t}^I$	$1 - \frac{O_{i,o,h,q,t}^I}{q} + \frac{G_{i,o,h,q,t}^I}{q}$	Fraction	pair.all.in.indirect.cost	Indirect quote-output loss across both legs, which includes both pools' fees and price impact, plus historical route gas as a fraction of input notional. Fee, price-impact, and gas contributions are retained separately and sum to this measure.
$\Delta C_{i,o,k,q,t}^{D,\text{all}}$	$C_{i,o,k,q,t}^{D^*} - C_{i,o,q,t}^{D^{\text{all}}}$	Fraction	pair.all.in.direct.cost.advantage	Indirect all-in cost minus direct all-in cost on common-support; positive values favor direct execution and negative values favor indirect execution.
$\text{DirectAvailable}_{k,t,q}$	$\frac{ \mathcal{D}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	direct.available.share	Fraction of pair $\text{pair} \in \mathcal{D}_{k,t,q}$ with $D_{i,o,q,t} = 1$. The unsuffixed data column fixes $q = \$10,000$.
$\text{IndirectAvailable}_{k,t,q}$	$\frac{ \mathcal{Z}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	vehicle.available.share	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $I_{i,o,h,q,t} = 1$, so both legs through k are executable.
$\text{IndirectOnlyAvailable}_{k,t,q}$	$\frac{ \mathcal{Z}_{k,t,q} \setminus \mathcal{D}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	no.direct.vehicle.available.share	Fraction of pairs with $D_{i,o,q,t} = 0$ and $I_{i,o,h,q,t} = 1$; no executable direct route but an executable route through k .
$\text{DirectDepth}_{k,t,q}$	$\text{median}_{\mathcal{D}_{k,t,q}} (O_{i,o,q,t}^D / q)$	Output USD per input USD	direct.depth.median	Median direct-route output ratio $O_{i,o,q,t}^D / q$ over executable-direct pairs $(i, o) \in \mathcal{D}_{k,t,q}$.
$\text{DirectCostAdvantage}_{k,t,q}$	$\text{median}_{\mathcal{C}_{k,t,q}} \Delta C_{i,o,k,q,t}^{D^*}$	Fraction	direct.cost.advantage.median	Median $\Delta C_{i,o,k,q,t}^{D^*}$ over common-support pairs $(i, o) \in \mathcal{C}_{k,t,q}$; positive values favor the direct route and negative values favor the indirect route through k .
$\text{AllInDirectCost$				